

Who Drops Out? A Multivariate Analysis of Academic Success in Portuguese Higher Education

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1. Introduction

Education is often described as a primary pathway for upward social and economic mobility. In principle, it provides individuals with the skills and opportunities needed to succeed based on merit. In practice, however, this process is not always straightforward. Many capable students still struggle or leave higher education, not necessarily because of a lack of ability, but because the system does not consistently identify or address challenges early enough.

One limitation is that support mechanisms such as tutoring, financial aid, or mental health resources are often introduced only after students begin to fall behind. At that stage, recovery becomes more difficult. This raises an important question. Rather than reacting to poor performance, can institutions detect early indicators of risk and intervene sooner?

This study focuses on identifying the key factors associated with dropout and evaluating whether these outcomes can be predicted early in a student's academic trajectory. Although the dataset is drawn from Portuguese higher education, the broader objective is to understand whether early academic and demographic signals can be used to inform more timely and targeted interventions. The results suggest that much of the divergence in student outcomes is already visible within the first year, which has important implications for how institutions allocate support.

2. Design and Primary Questions

The primary goal of this study is to understand the key factors that influence student dropout and evaluate whether these outcomes can be predicted early in a student's academic path. To approach this, the analysis follows a sequence of multivariate techniques guided by central questions: what factors contribute most to student dropout, how accurately can dropout be predicted, and which variables are most influential in that process. In addition, the study considers broader patterns of success, including whether certain groups of students consistently perform better and what institutional mechanisms, such as scholarship programs, may be effective in supporting student outcomes.

First, Principal Components Analysis (PCA) was used to summarize the data and identify the main underlying patterns. A focus was on the consolidation of the 40 available features into a few recognizable patterns, perhaps across academic ability and demographics, and to see if any outliers shared characteristics. The focus here is on what patterns are present in the data and whether the large number of variables can be reduced into meaningful groups that explain student outcomes.

Next, Multivariate Analysis of Variance (MANOVA) was applied to test for differences between groups, attempting to answer the main questions of whether, when controlling for ability (performance before college), demographic identities like being a displaced student, a woman, or being in debt to the university are contributing factors to graduation. This is where the focus on categorical variables is most at play, examining whether different groups perform differently and which demographic or institutional factors significantly impact outcomes.

Finally, Discriminant Analysis (DA) was then used to assess how well students are classified into outcomes and to identify which variables contribute most strongly to the classification, focusing on how accurately student outcomes can be predicted and which variables are most important in distinguishing between groups.

3. Data

3.1. Dataset Description

The dataset used in this analysis is a consolidation of multiple administrative sources collected from higher education institutions in Portugal as part of a government-supported initiative aimed at improving academic outcomes. While the data is drawn from Portuguese schools, it includes a diverse range of 4424 students from a range of European and Latin American countries. The dataset captures 40 dimensions of the student experience, including background characteristics, family and socioeconomic context, enrollment details, and academic performance during the first year, allowing for a comprehensive view of both pre-enrollment factors and early academic progression.

3.2. Variables

Table 1: Dataset Variables

Variable Name	Type	Description
Student Background		
marital_status	Categorical	Marital status at enrollment (single, married, etc.)
nacionality	Categorical	Country of nationality (Portuguese, Brazilian, etc)
displaced	Binary	Moved away from home (1 = yes, 0 = no)
educational_special_needs	Binary	Declared special needs (1 = yes, 0 = no)
gender	Binary	Gender (1 = male, 0 = female)
international	Binary	International student (1 = yes, 0 = no)
Family Background		
mothers_qualification	Categorical	Mother's education level (cannot read to doctorate)
fathers_qualification	Categorical	Father's education level (cannot read to doctorate)
mothers_occupation	Categorical	Mother's job category (International Codes)
fathers_occupation	Categorical	Father's job category (International Codes)
Enrollment Details		
application_mode	Categorical	Application type (general, transfer, over-23, etc)
application_order	Continuous	Course preference ranking (0 = first choice, 9 = last)
course	Categorical	Degree program (nursing, management, etc)
daytime_evening_attendance	Binary	Attendance type (1 = daytime, 0 = evening)
previous_qualification	Categorical	Prior education (secondary, bachelor's, etc.)
previous_qualification_grade	Continuous	Prior grade (0–200 scale)
admission_grade	Continuous	Admission grade (0–200 scale)
debtor	Binary	Owes past tuition (1 = yes, 0 = no)
tuition_fees_up_to_date	Binary	Fees paid on time (1 = yes, 0 = no)
scholarship_holder	Binary	Has scholarship (1 = yes, 0 = no)
age_at_enrollment	Continuous	Age at enrollment (years)
1st Year Performance		
curricular_units_1st_sem_credited	Continuous	Transfer credits (semester 1)
curricular_units_1st_sem_enrolled	Continuous	Courses enrolled (semester 1)
curricular_units_1st_sem_evaluations	Continuous	Exams taken (semester 1, incl. resits)
curricular_units_1st_sem_approved	Continuous	Courses passed (semester 1)
curricular_units_1st_sem_grade	Continuous	Avg grade (0–20 scale; 0 if none passed)
curricular_units_1st_sem_without_evaluations	Continuous	Not assessed courses (semester 1)
curricular_units_2nd_sem_credited	Continuous	Transfer credits (semester 2)
curricular_units_2nd_sem_enrolled	Continuous	Courses enrolled (semester 2)
curricular_units_2nd_sem_evaluations	Continuous	Exams taken (semester 2)
curricular_units_2nd_sem_approved	Continuous	Courses passed (semester 2)
curricular_units_2nd_sem_grade	Continuous	Avg grade (0–20 scale)
curricular_units_2nd_sem_without_evaluations	Continuous	Not assessed courses (semester 2)
National Markers		
unemployment_rate	Continuous	Unemployment rate (%) at enrollment
inflation_rate	Continuous	Inflation rate (%), can be negative)
gdp	Continuous	GDP growth rate (can be negative)
Graduation Status		
target	Categorical	Outcome (Dropout, Enrolled, Graduate)

3.3. Data Quality & Limitations

Before analysis, the data was cleaned to make sure everything was consistent across sources by the researchers who created the dataset. During my initial exploration, a few anomalies showed up, including observations that didn't really match typical enrollment patterns, like a 30 year old student applying through a standard first-round admission process. These cases were pretty rare, but they could point to data entry issues or just edge cases in the population, and they highlight some limitations in the dataset. This was kept these in mind during the analysis, trying to reduce their impact while still preserving meaningful variation in the data.

4. Descriptive Plots and Summary Statistics

4.1. Summary Statistics

Table 2: Student Summary Statistics

Metric	Total	Percentage
Total students	4424	100%
Graduate	2209	49.9%
Dropout	1421	32.1%
Enrolled	794	17.9%
Female	2868	64.8%
Male	1556	35.2%
Fees up to date	3896	88.1%
Scholarship holders	1099	24.8%
Debtors	503	11.4%
International	110	2.5%
Displaced	2426	54.8%

Table 3: Top 3 Majors, Admissions Types, and Nationalities

Top 3 Majors

- 1 Nursing (766)
- 2 Management (380)
- 3 Social Service (355)

Top 3 Admission Types

- 1 1st phase - general contingent (1708)
- 2 2nd phase - general contingent (872)
- 3 Over 23 years old (785)

Top 3 Nationalities

- 1 Portuguese (4314)
- 2 Brazilian (38)
- 3 Santomean (14)

4.2. Descriptive Plots

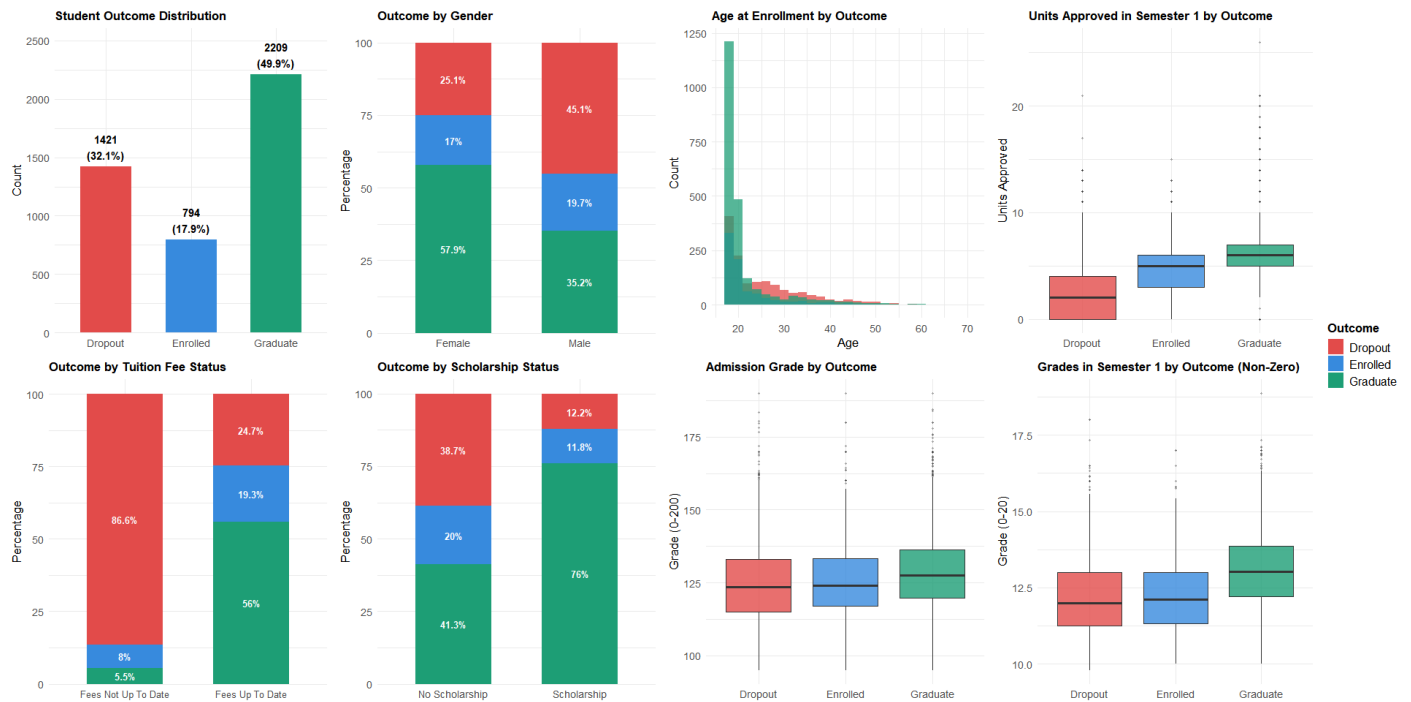


Figure 1: Bar Charts and Box Plots of Demographic Data By Outcome

4.3. Notes on the Portuguese Education System

In the Portuguese higher education system, grades are typically on a 0–20 scale, where a 10 is considered passing and anything above that reflects stronger performance. Admissions happen in multiple rounds (like 1st phase, 2nd phase), where the first round fills most spots based on admission grades, and later rounds give students another chance to get placed if seats are still open. It’s also important to note that just being enrolled in a course doesn’t mean you earn credit. You usually have to actually sit for and pass the exam. If you don’t take the exam or don’t pass, you don’t get credit, even if you were technically enrolled in the class.

5. Multivariate Analysis and Discussion

5.1. Principal Component Analysis

5.1.1. Evaluation of Normality And Correlations

When evaluating the continuous variables, there was multiple violations of normality. Several variables were zero inflation, especially units credited (transfer credits) because the dataset contains relatively few transfer students. Additionally, some variables required scaling for meaningful comparison, such as inflation rate, since it can take negative values.

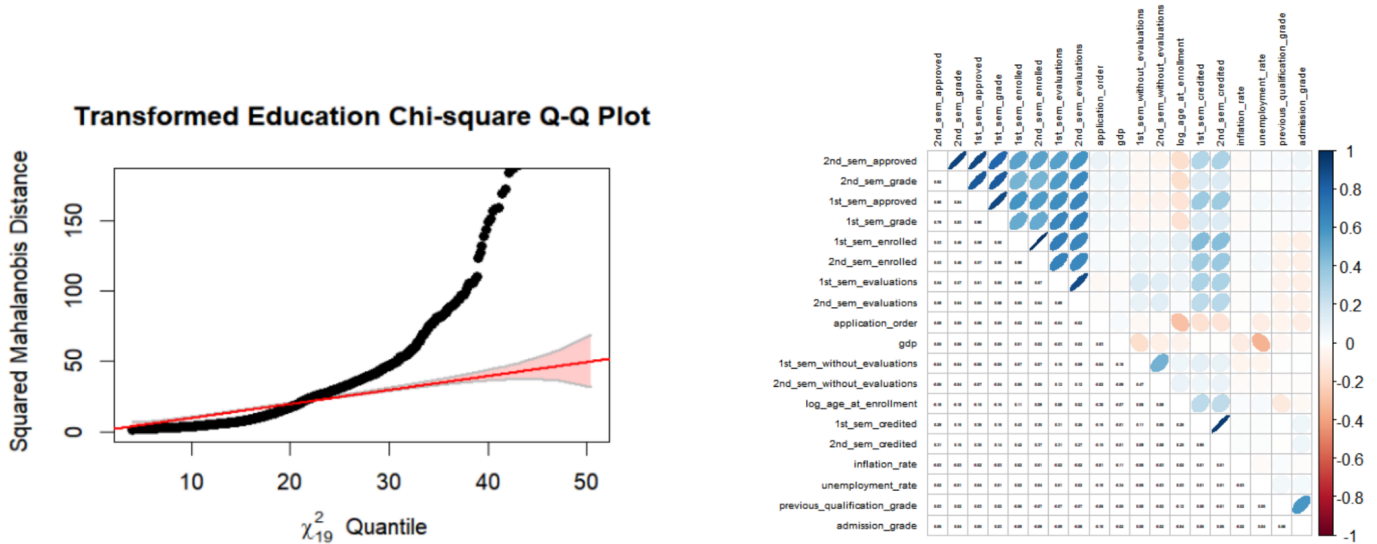
Despite applying a series of transformations, including log, square root, and scaling transformations, the resulting chi-squared plot still indicated a lack of multivariate normality. This outcome was expected given the large presence of count-based variables in the dataset. Since PCA does not require multivariate normality, I proceeded with the analysis.

Strong positive correlations are observed between first and second semester units approved, grades, and enrolled units. There is also a clear relationship between first and second semester credited units, as well as between previous qualification grades and admission grades.

These relationships are intuitive. Students enrolled in more courses tend to pass more courses, and prior academic performance naturally aligns with admission outcomes. Negative correlations are generally weaker but still present. Notably, application order is negatively correlated with age at enrollment, suggesting that older students are more likely to enroll in programs closer to their preferred choice. GDP and unemployment rate also show a negative relationship, which aligns with general economic expectations.

Some variables, such as credited units, courses without evaluations, and national indicators, exhibit weaker correlations with the rest of the dataset. This aligns with the low prevalence of transfer students and the limited variation in national-level factors, given that most students are from Portugal.

Given this correlation structure, PCA is well suited for dimensionality reduction. The next step is determining how many principal components to retain.



(a) Chi-squared plot used to assess multivariate normality.

(b) Correlation matrix of continuous variables.

Figure 2: Normality assessment and correlation structure of the dataset.

5.1.2. Principle Components Analysis

Initial inspection of the scree plot suggests that the first two or three principal components capture a substantial portion of the variance.

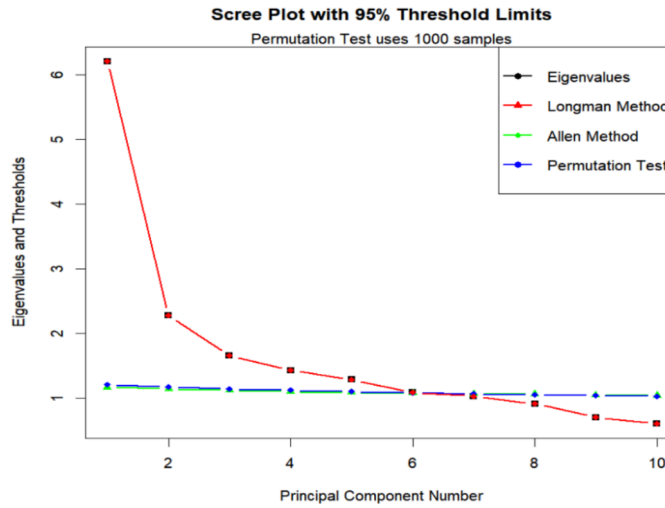


Figure 3: Scree plot of eigenvalues by each principal component.

However, when compared with the Longman method, Allen method, and permutation testing, the results indicate that retaining six principal components is more appropriate.

5.1.3. Interpretation of Principal Components

To evaluate the effectiveness of dimensionality reduction, score plots were analyzed.

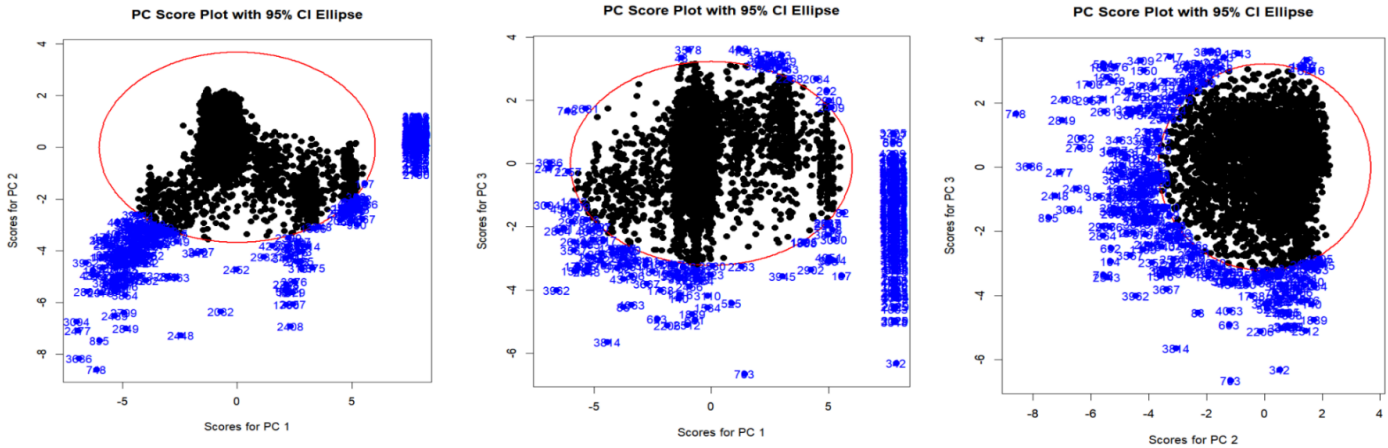


Figure 4: Pairwise score plots of the first 3 Principal Components.

Several clusters appear as outliers, particularly observations with PC1 values exceeding 6 and a distinct group in the lower-left quadrant where PC1 is less than -2.5 and PC2 is less than -2.5.

Further investigation shows that observations with PC1 values above 6 correspond to students with zero values across enrollment-related variables, including enrolled, approved, and credited units. These likely represent students who effectively dropped out but have not been formally recorded as such.

The cluster in the lower-left quadrant provides a more compelling insight. Summary statistics indicate that this group has an average age of approximately 32. Examination of application mode shows that these students are predominantly non-traditional, including transfer students or individuals returning to education after prior study. This highlights that PCA is not only reducing dimensionality but also uncovering distinct subpopulations within the data.

Table 4: Top Loadings for Principal Components (PC1–PC3)

Rank	PC1	PC2	PC3
1	1st sem approved (-0.37)	1st sem credited (-0.43)	admission grade (-0.62)
2	2nd sem approved (-0.35)	2nd sem credited (-0.43)	previous qualification grade (-0.59)
3	1st sem evaluations (-0.34)	age at enrollment (-0.42)	
4	1st sem grade (-0.34)	1st sem without evaluations (-0.26)	
5	2nd sem evaluations (-0.34)		

Table 5: Top Loadings for Principal Components (PC4–PC6)

Rank	PC4	PC5	PC6
1	1st sem without evaluations (-0.54)	unemployment rate (0.64)	application order (0.62)
2	2nd sem without evaluations (-0.52)	gdp (-0.53)	inflation rate (0.44)
3	gdp (0.39)	2nd sem without evaluations (-0.28)	age at enrollment (-0.36)
4	1st sem credited (0.26)	1st sem without evaluations (-0.27)	2nd sem credited (0.27)
5	2nd sem credited (0.26)		

To determine how many principal components to retain, interpretability of the loadings was considered, focusing on values with magnitude greater than 0.25.

The first principal component captures overall academic performance, with strong contributions from variables such as courses passed, evaluations completed, and grades during early semesters. The second component reflects non-traditional student characteristics, driven primarily by credited units and age at enrollment. The third component represents pre-college academic ability, with strong loadings on admission grades and prior academic performance. The fourth component is less directly interpretable but appears to capture patterns related to courses where students were enrolled but did not complete evaluations. The fifth component reflects broader national indicators, such as unemployment rate and GDP. The sixth component is less interpretable but may again relate to non-traditional pathways, incorporating variables such as age and application order.

Notably, it is only after including six principal components that over 70 percent of the total variance is explained. This indicates that all six components contribute meaningfully to the structure of the data and should be retained for further analysis.

5.2. MANOVA

5.2.1. Baseline MANOVA Model

$$\mathbf{Y} = \begin{bmatrix} \text{Sem1 Enrolled} \\ \text{Sem1 Approved} \\ \text{Sem1 Grade} \\ \text{Sem2 Enrolled} \\ \text{Sem2 Approved} \\ \text{Sem2 Grade} \end{bmatrix} = \beta_0 + \beta_1 \text{Target} + \beta_2 \text{Fees} + \beta_3 (\text{Target} \times \text{Fees}) + \varepsilon$$

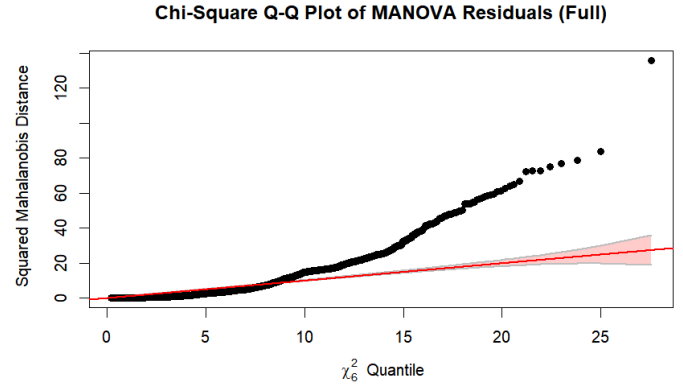
The two-way MANOVA evaluates whether graduation status and tuition fee status predict academic performance in the first year. Outcome group is the dominant factor in explaining performance. The multivariate effect is strong (Pillai ≈ 0.171 , $F \approx 68.8$, $p < 2.22 \times 10^{-16}$), and this is reflected across all outcomes, with very large F-statistics, including approximately 260 for semester 1 grades and 416 for semester 2 approved units. Dropouts, enrolled students, and graduates occupy clearly distinct academic profiles.

Tuition fee status plays a much smaller role. Although it is statistically significant at the multivariate level (Pillai ≈ 0.0042 , $p \approx 0.0046$), it is not significant for any individual performance measure, with p-values generally above 0.1. The interaction between outcome group and fee status is also weak overall ($p \approx 0.10$), with only a marginal effect for first-semester approved units ($p \approx 0.046$). This suggests that fee status does not meaningfully explain differences in academic performance once students are grouped by outcome.

Table 6: Academic Performance by Tuition Fee Status and Outcome in Base MANOVA Model

Outcome	n	Mean Approved	Mean Grade	% Zero
Fees Not Up to Date				
Dropout	457	2.26	6.70	44.4
Enrolled	42	4.90	11.47	11.9
Graduate	29	6.31	11.51	10.3
Fees Up to Date				
Dropout	964	2.69	7.52	38.1
Enrolled	752	4.29	11.11	8.8
Graduate	2180	6.23	12.66	3.4

Figure 5: Chi-squared of Residuals



Overall, the separation between outcome groups is large meaningful, while fee status adds very little additional explanatory power. This points to the idea that dropout is not primarily driven by differences in grades or immediate financial standing, but by broader factors that emerge during enrollment, such as engagement, institutional support, and overall student circumstances.

The residuals deviate from normality, which is expected given the count-based and zero-inflated nature of many variables, along with clear subgroup structure in the data. This suggests differing covariance structures across groups, motivating more flexible modeling approaches.

5.2.2. Model Controlling for Pre-Admission Performance

I wanted to see if controlling for grades before college, tuition status and graduation status could still predict college academic performance.

$$\mathbf{Y} = \begin{bmatrix} \text{S1 Enr} \\ \text{S1 App} \\ \text{S1 Grade} \\ \text{S2 Enr} \\ \text{S2 App} \\ \text{S2 Grade} \end{bmatrix} = \beta_0 + \beta_1 \text{Target} + \beta_2 \text{Fees} + \beta_3 (\text{Target} \times \text{Fees}) + \beta_4 \text{Admission Grade} + \beta_5 \text{Prior Qualification Grade} + \varepsilon$$

Table 7: Regression Coefficients for Semester 1 Grade

Term	Estimate	SE	t	p	Sig
targetEnrolled	4.742	0.679	6.988	3.21e-12	***
targetGraduate	4.741	0.808	5.870	4.68e-09	***
tuition_fees_up_to_dateyes	0.805	0.239	3.368	7.65e-04	***
admission_grade	0.006	0.005	1.065	2.87e-01	
previous_qualification_grade	0.001	0.006	0.117	9.07e-01	

Graduates earn roughly 4.7 points higher grades in the first semester than dropouts, with enrolled students showing a similarly strong but slightly smaller advantage. This supports the interpretation that enrolled students are closer to delayed graduates rather than a distinct at-risk group. Interestingly, having tuition fees up to date contributed to higher grades (a whole 0.805 points on a 20 point scale) in college after controlling for ability before college. Importantly, neither admission grade nor previous qualification grade significantly predicts first-semester grades, indicating that these differences are not driven by prior academic ability. Tuition fee status provides a smaller but consistent advantage, with students who are up to date on fees earning higher grades.

5.2.3. Multivariate Contrasts: Drivers of Graduation

Since we analyzed if graduation status could predict performance, I now wanted to see what factors predicted graduation status. The MANOVA contrast results compare two setups. One predicts graduation directly from socioeconomic factors, while the other controls for academic performance. The first shows total effects, and the second shows what remains when students have the same first year grades.

Financial variables play a major role. Being behind on fees is significant as a predictor for graduation before and after controlling for grades in the first year. This means the impact is not just academic but affects whether students can remain enrolled. Prior tuition debt shows a similar pattern, remaining stable across models. Scholarship status is the strongest factor, with students graduating at higher rates even when academic performance is the same, highlighting the effectiveness of preventative support.

Other variables show similar patterns. Women graduate at higher rates than men even with identical performance, suggesting non-academic differences. Displacement becomes more significant after controlling for grades, indicating structural barriers rather than academic ones. International status and attendance type are not significant.

Table 8: F(P) MANOVA Contrasts for Graduation Outcomes

Question	Without Performance	With Performance
Female vs Male	110.1***	22.4***
Fees Up To Date	249.4***	76.1***
Scholarship	282.8***	180.8***
Prior Debt	38.6***	34.7***
International	1.06	1.54
Evening vs Day	1.33	0.56
Displaced	5.61*	6.66**
Admission Grade	32.9***	24.4***

Overall, graduation outcomes are not primarily driven by academic ability. Instead, they reflect external factors that influence whether students remain enrolled and complete their degree. Financial stability and institutional support play a central role, and the strength of the scholarship effect highlights how targeted interventions can directly improve academic success.

5.3. Discriminant Analysis

5.3.1. Checking Assumptions

The assumptions required for linear discriminant analysis were not satisfied. Box's M test strongly rejected the equality of covariance matrices across outcome groups ($p = 2.22e-16$), indicating substantial differences in group-specific variance structure. In addition, intragroup normality diagnostics deviated heavily from multivariate normality in all 3 groups. Because LDA assumes both equal covariance matrices and approximate normality, these violations suggest that a linear decision boundary is not appropriate. Instead, quadratic discriminant analysis (QDA) was used, allowing for group-specific covariance structures and a more flexible, nonlinear separation that better aligns with the underlying data.

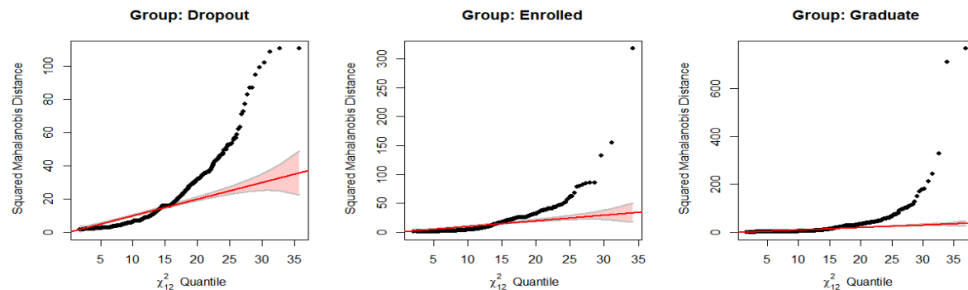


Figure 6: Chi-Squared QQ-Plots to Check Normality Between Graduation Groups

5.3.2. Results

To classify students into dropout, enrolled, and graduate groups, quadratic discriminant analysis (QDA) was used. Before fitting the model, Wilks' Lambda was used to confirm that outcome groups differ jointly across the selected predictors. The test indicated strong multivariate separation, reinforcing the earlier MANOVA results and justifying the use of discriminant analysis.

Raw QDA (Accuracy = 0.697)			
	Dropout	Enrolled	Graduate
Dropout	740	371	310
Enrolled	74	334	386
Graduate	26	175	2008
CV QDA (Accuracy = 0.693)			
	Dropout	Enrolled	Graduate
Dropout	738	372	311
Enrolled	77	328	389
Graduate	27	181	2001

Table 9: Confusion Matrix for QDA Results

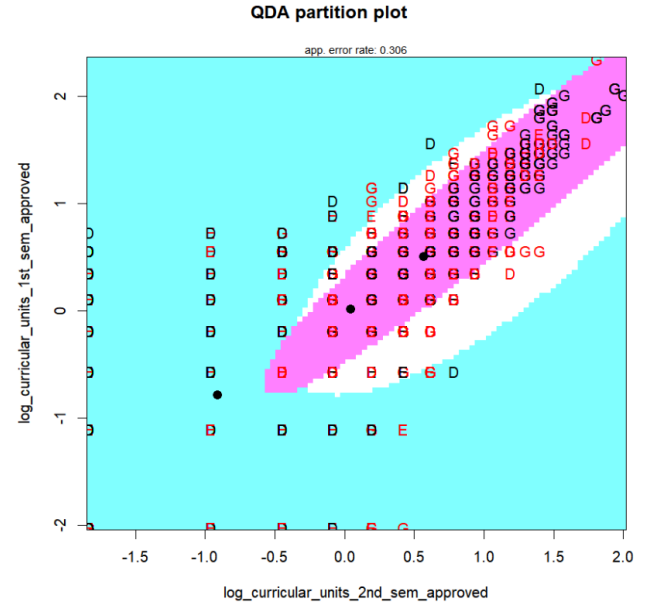


Figure 7: Discriminant Analysis Partition Plot

The QDA model achieves strong classification performance, with an overall accuracy of 69.7% and cross-validated accuracy of 69.3%, both substantially higher than the 33.3% chance level. The stability between raw and cross-validated accuracy suggests that the model generalizes well and is not overfitting.

Examining the confusion matrix reveals that classification performance is strongest for graduates, with the majority of cases correctly identified. Dropouts are also classified reasonably well, though some confusion occurs with enrolled students. The largest source of misclassification is within the enrolled group, which is frequently predicted as either dropout or graduate. This pattern aligns with earlier findings and suggests that enrolled students do not form a clearly distinct category.

The misclassification of enrolled students highlights the heterogeneity within this group. Some enrolled students are likely delayed graduates, temporarily paused due to factors such as medical leave or personal circumstances that extend their academic timeline without reflecting poor performance. Others may be progressing more slowly due to weaker academic alignment or disengagement, placing them closer to the dropout trajectory. This mixture explains why enrolled students are often classified into both neighboring groups, as they represent both delayed success and emerging risk.

To visualize how the model is actually separating the groups, a QDA partition plot was created using 1st and 2nd semester passed courses. The decision boundaries are clearly curved, which reinforces that a linear model wouldn't have been a good fit here. There's a clear progression from dropout to graduate as approved units increase. Graduates cluster tightly in the high-performance region (purple), while dropouts sit on the lower end (blue). Enrolled students fall right in the middle and overlap with both groups. This lines up with the confusion matrix, where enrolled students are the ones most often misclassified, since they don't form a clean, separate group.

Overall, the results confirm that student outcome groups are highly separable based on academic performance within the first 2 semesters of university, with most classification errors occurring at the boundary between enrolled students and the other groups rather than between dropouts and graduates.

The dataset shows a moderate class imbalance, with graduates comprising nearly half of all observations, while enrolled students represent the smallest group. This imbalance can influence classification performance, as models tend to favor the majority class, contributing to the high accuracy observed for graduates. In contrast, the enrolled group is both smaller and more heterogeneous, making it more prone to misclassification. As a result, overall accuracy should be interpreted with some caution, as it reflects weaker performance on the more ambiguous minority group.

Table 10: k-NN Classification Accuracy by Number of Neighbors

k	Accuracy
3	0.678
5	0.694
15	0.705
25	0.706
50	0.694

Since the data suggested non-linear relationships, I decided to also run k-nearest neighbors analysis to test on no parametric assumptions.

The k-NN results show a clear improvement in classification accuracy as the number of neighbors increases, rising from 0.678 at k=3 to a peak of 0.706 at k=25, before slightly declining at higher values. This pattern reflects the typical bias–variance tradeoff, where small values of ks are more sensitive to noise, while larger values smooth over local variation. The optimal performance at intermediate ks suggests that student outcomes are not driven by highly localized patterns, but instead emerge from broader similarity across performance profiles.

Importantly, the overall accuracy remains consistently high and well above chance (0.33), and is comparable to the QDA results. This indicates that both local (k-NN) and nonlinear global (QDA) approaches are able to capture the underlying structure in the data. At the same time, the relatively small gains across different values of k suggest that the primary challenge is not separating clear groups like graduates and dropouts, but rather handling the overlap associated with the enrolled group, which lies between the two extremes.

6. Conclusions and Discussion

The results across PCA, MANOVA, and classification models point to a consistent pattern. Student outcomes begin to diverge very early, and much of that separation is already visible within the first year. Academic performance, especially courses approved and grades, is the strongest driver of this separation. This is reflected in the MANOVA results, where outcome group shows extremely large effects, with F-statistics reaching approximately 260 for first semester grades and 416 for second semester approved units.

At the same time, the analysis shows that performance alone does not fully explain outcomes. When shifting the focus to predicting graduation, financial and institutional factors become more important. Tuition fee status remains significant even after controlling for performance ($F = 76.1$), and prior debt shows only a small change ($F = 38.6$ to 34.7), suggesting that these effects operate independently of academic ability. Most notably, scholarship status remains the strongest and most stable factor ($F = 282.8$ to 180.8), indicating that support systems play a direct role in helping students persist and graduate.

The classification results reinforce this structure. QDA achieves an accuracy of 69.7% (69.3% cross-validated), well above the 33.3% chance level, and k-NN reaches a similar peak of 0.706 at $k = 25$. These results show that student outcomes are highly predictable based on first-year performance alone. However, the main difficulty in classification is not between dropouts and graduates, which are well separated, but within the enrolled group. Enrolled students tend to fall between the two and are frequently misclassified into both neighboring groups, reflecting that they represent a mix of delayed success and emerging risk rather than a clearly defined category.

These results suggest that dropout is not primarily driven by academic ability. Pre-admission measures such as admission grade and prior qualification grade are largely insignificant in predicting early performance, while outcomes are more strongly shaped by what happens after enrollment. This points to the influence of structural and external factors, including financial stability, access to support, and broader life circumstances.

From an institutional perspective, this highlights a gap in how student support is currently implemented. Current approaches tend to rely on performance-based indicators, but this analysis suggests that by the time performance declines, underlying issues are already present. A more effective approach would focus on early identification and targeted support. This could include expanding scholarship programs, given their strong and consistent effect, as well as providing proactive financial assistance to students at risk of falling behind on fees. Institutions may also need to pay closer attention to groups such as displaced students, who perform adequately but still face barriers to graduation.

More broadly, the findings suggest that education systems would benefit from shifting toward a more preventative

approach. Instead of waiting for students to fail courses, institutions can use early performance data and demographic indicators to identify at-risk students and intervene before disengagement occurs. This may involve structured mentoring programs, mental health support, and more flexible academic pathways for non-traditional students. Improving graduation outcomes is therefore not only about academic performance, but also about addressing the conditions that affect whether students are able to remain enrolled and complete their degrees.

7. Points for Further Analysis

While this analysis provides strong evidence that student outcomes are predictable and influenced by structural factors, there are several directions for further work that could extend these findings.

First, the enrolled group warrants more detailed investigation. The classification results show that this group is highly heterogeneous, with students distributed across both dropout and graduate trajectories. Applying clustering techniques within this group could help identify distinct subtypes, such as delayed graduates versus at-risk students, allowing for more targeted interventions rather than treating all enrolled students as a single category.

Second, while this study highlights the importance of financial and institutional factors, it does not directly measure variables such as mental health, student engagement, or access to support services. Incorporating these types of variables could provide additional insight into why factors like scholarship status and displacement remain significant even after controlling for academic performance.

Additionally, alternative modeling approaches could be explored. While QDA and k-NN perform well, more flexible machine learning models such as random forests or gradient boosting may improve classification accuracy and provide more detailed measures of variable importance. These models may also capture nonlinear relationships that are not fully represented in the current analysis. The original dataset includes some machine learning analysis, suggesting this would be a natural extension.

Finally, the dataset is specific to Portuguese higher education, and while it includes some international students, the findings may not fully generalize to other systems. Replicating this analysis using data from different countries or institutions would help determine whether these patterns are consistent or context-dependent.

Overall, further analysis should focus on improving the identification of at-risk students and better understanding the mechanisms behind structural barriers to graduation. Combining more detailed data with more flexible modeling approaches may help move toward systems that not only predict student outcomes, but also support improved outcomes over time.